Topology as a Computational Basis for ICL in First-Order CRNs
A concise account of the theory, experiments, audit, and current evidence

Topology worktree: `topology`
Evidence source commit: `3c9c685`
Markov extension: local artifacts
May 8, 2026

Summary. For first-order CRNs with exponential input-dependent rates, topology is not just a container for trainable parameters. It determines the rooted-spanning-tree sums through which edge projections enter the steady state. The computational basis is therefore not the set of edge vectors $\{K_e\}$, but the set of tree-sum vectors

$$\boxed{\Theta_T = \sum_{e \in T} K_e, \qquad T \in \mathcal{T}_r(G),\ r \in V.}$$

The experiments support the scoped statement that, in the tested fixed-count regimes, topology-associated structural and functional variables explain residual novel-class ICL variation that raw trainable count does not explain. They do not yet give a universal scalar topology law.

Scope. Everything below is about first-order CRNs with exponential rate encoding. The matrix-tree theorem makes this case exact. Autocatalytic and WTA CRNs need separate topology theories.

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[1] The starting question

The original CRN-ICL paper showed that a chemical network can do in-context learning without transformer attention. The input is a context/query vector

$$z = (z_1, \ldots, z_{N_c}, z_q), \qquad z_i \in \mathbb{R}^D,$$

and the target is the context position whose item matches the query. The true metric is novel-class ICL accuracy, not training accuracy or ordinary validation accuracy.

The paper's mechanism was subspace projection. For each context position i, define a comparison region

$$\mathcal{M}_i = \{z : z_i \sim z_q\}.$$

ICL succeeds when inputs from different comparison branches produce steady states that the decoder separates.

The original degree-count story was broadly right:

$$n_{\mathrm{req}} = 2N_c(N_c + 1)D, \qquad n_g = N_n(N_n - 1)(N_c + 1)D$$

for fully connected first-order CRNs. But this count does not say where the degrees of freedom sit. The topology question was therefore

$$\boxed{\text{At fixed count, does the reaction/input topology change ICL?}}$$

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[2] The exact first-order object

Let $G = (V, E)$ be a strongly connected directed first-order reaction graph. For edge $e : u \to v$, use exponential rate encoding

$$k_e(z) = \exp(b_e + K_e^\top z).$$

The first-order steady state is controlled exactly by the matrix-tree theorem:

$$\bar{C}_r(z) = \frac{\tau_r(z)}{\sum_s \tau_s(z)}, \qquad \tau_r(z) = \sum_{T \in \mathcal{T}_r(G)} \prod_{e \in T} k_e(z).$$

Substitute the exponential form:

$$\prod_{e \in T} k_e(z) = \exp\left(\sum_{e \in T} b_e + \left(\sum_{e \in T} K_e \right)^\top z \right).$$

Set

$$\beta_T = \sum_{e \in T} b_e, \qquad \Theta_T = \sum_{e \in T} K_e.$$

Then

$$\boxed{\tau_r(z) = \sum_{T \in \mathcal{T}_r(G)} \exp(\beta_T + \Theta_T^\top z).}$$

This is the key point. A first-order CRN does not read out isolated edge projections $K_e^\top z$. It reads out log-sum-exp combinations of rooted-tree projections $\Theta_T^\top z$.

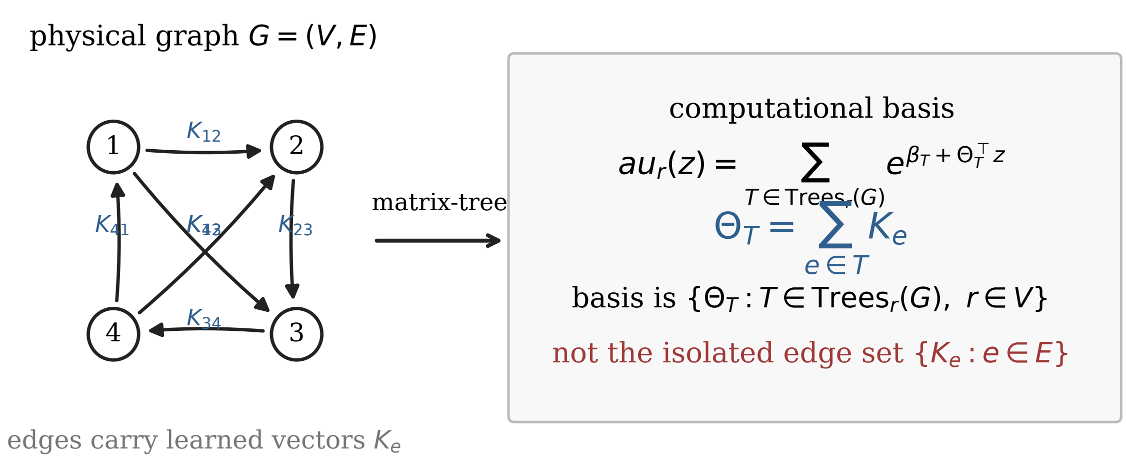

Figure 1: Topology changes the projection basis. Edge vectors K_e are learned locally, but the steady state uses rooted tree-sum projections Θ_T.

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[3] What topology changes

Stack tree incidences as row vectors $s_T \in \{0,1\}^m$. Then

$$\Theta_T = K^\top s_T.$$

Since concentrations are normalized, common shifts of all tree scores cancel. The more relevant structural object is the tree-difference matrix

$$D_G = \begin{bmatrix} s_{T_1} - s_{T_0} \\ s_{T_2} - s_{T_0} \\ \vdots \end{bmatrix}.$$

A first count of relative projection capacity is

$$d_{\text{rel}}(G) = p\ \text{rank}(D_G), \qquad p = (N_c + 1)D.$$

With an input mask $\Omega_{e\alpha}$, the capacity must be input-aware:

$$\boxed{d_{\text{rel}}(G,\Omega) = \sum_{\alpha=1}^{p} \text{rank}\left(D_G\,\text{diag}(\Omega_{\cdot\alpha})\right).}$$

But this is still only a rank proxy. ICL asks a sharper question:

$$\boxed{\text{Can the allowed tree-sum directions separate the comparison branches?}}$$

The capacity object should be closer to

$$\gamma^*(G,\Omega) = \max_{K,B} \min_{z\in\mathcal{B}} \left[q_{\ell^*(z)}(z) - \max_{\ell\neq\ell^*(z)} q_\ell(z) \right]$$

under norm constraints. In the tropical limit,

$$\log \tau_r(z) \approx \max_{T\in\mathcal{T}_r(G)} (\beta_T + \Theta_T^\top z),$$

so the theory becomes a rooted-tree-polytope normal-fan coverage problem.

The hierarchy that emerged is therefore

$$\begin{aligned} &\text{raw count} < \text{relative tree rank} < \text{masked tree geometry} \\ &\qquad < \text{branch-margin/tree-polytope geometry} < \text{trained organization} < \text{causal dependence.} \end{aligned}$$

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[4] What was built and audited

The worktree now contains code and reports for arbitrary first-order directed topologies, input masks, tree metrics, branch-capacity probes, post-training tree diagnostics, ablations, motif extraction, and statistic-preserving causal scrambles.

The audit report `ICL/results/next_phase_stats/topology_goal_completion_audit.md` maps the goal to concrete artifacts. It records that the verifier passed for committed artifacts, that hard regimes have full result/model/mechanism/causal follow-ups, and that all headline claims are scoped to `test_novel_classes`. The audit also marks the branch-margin theory as partial by design: proxies exist, but there is not yet a final theorem.

The main source reports used here are:

`topology_icl_complete_explanation.md`	full narrative explanation
`next_phase_evidence_report.md/json`	fixed-count, motif, causal summaries
`mechanism_isolation_evidence.csv`	controlled mechanism contrasts
`stat_preserving_causal_stratified`	stratified causal scrambles
`degree_rewire_normal_fan_...`	exact-degree normal-fan pilot

[5] Fixed-count structural evidence

The original controlled regime fixed

$$N_n = 6, \qquad m = 20, \qquad \text{input-coupled count} = 200.$$

It used three physical backbones, 16 selected masks/topologies per backbone, five training seeds each, 240 total runs, and 48 topology/mask groups.

At group level, with target equal to mean novel-class ICL accuracy, the leave-one-out values were

model	group LOO R^2	interpretation
raw count	−0.043	no explanatory power at fixed count
raw count+d_{rel}	0.145	relative rank helps
masked tree geometry	0.189	input-aware geometry helps
tree geometry	0.409	strongest structural summary here

The corresponding hard $N_n = 5, m = 12, N_c = 3, D = 2$ regime showed a different but still positive pattern: raw count and d_{rel} alone were both −0.190, tree geometry was 0.324, and masked tree geometry was 0.447.

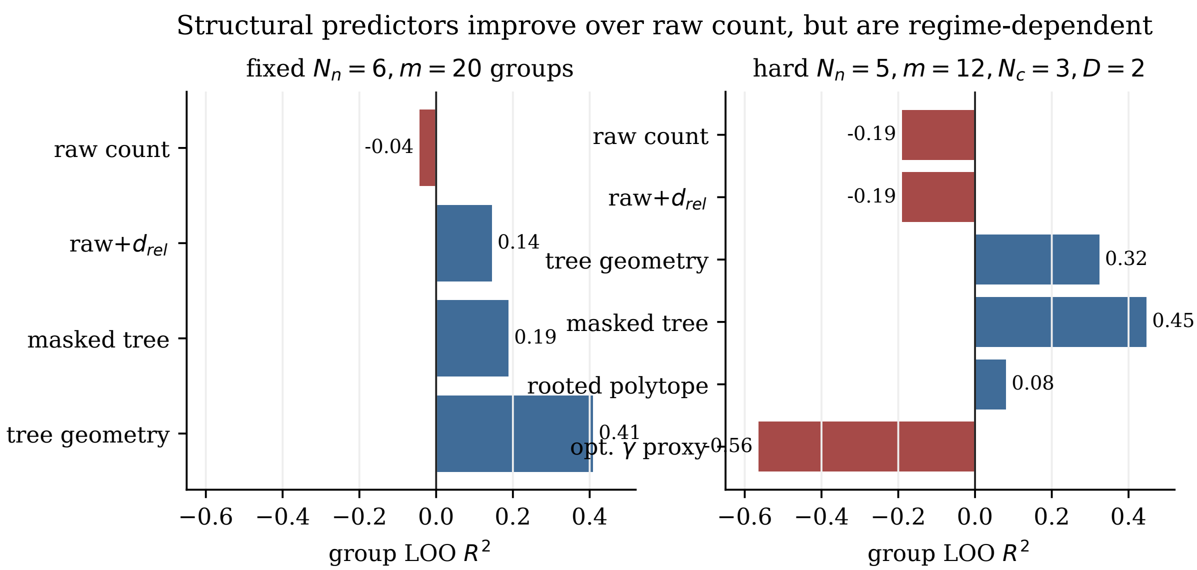

Figure 2: Structural predictors in two fixed-count regimes. Raw count is uninformative once fixed. Tree geometry and masked tree geometry explain residual group-level variation, but the best predictor depends on regime.

The safe conclusion is not that d_{rel} is the law. It is that topology-associated variables explain residual ICL variation that raw count cannot explain in the tested regimes.

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[6] Why input masks matter

Physical topology and input-encoding topology are distinct. The physical graph fixes the rooted trees:

$$G_{\mathrm{phys}} \mapsto \mathcal{T}_r(G).$$

The input mask fixes which coordinates can enter which edge vectors:

$$\Omega_{e\alpha} = 0 \quad \Rightarrow \quad K_{e\alpha} = 0.$$

The real capacity object is therefore $\gamma^*(G, \Omega)$, not only $\gamma^*(G)$.

The mechanism-isolation file shows fixed-graph, fixed-count, fixed-d_{rel} cases where masks still change ICL. On the fixed random physical graph, changing edge-load structure gave $+6.96$ mean ICL points, while changing coordinate-load heterogeneity gave -16.48 mean ICL points. On the fixed cycle graph, the corresponding deltas were -6.32 and $+2.12$ points; on the fixed hub graph they were $+3.44$ and -1.52 points. The sign is not universal, but the effect exists.

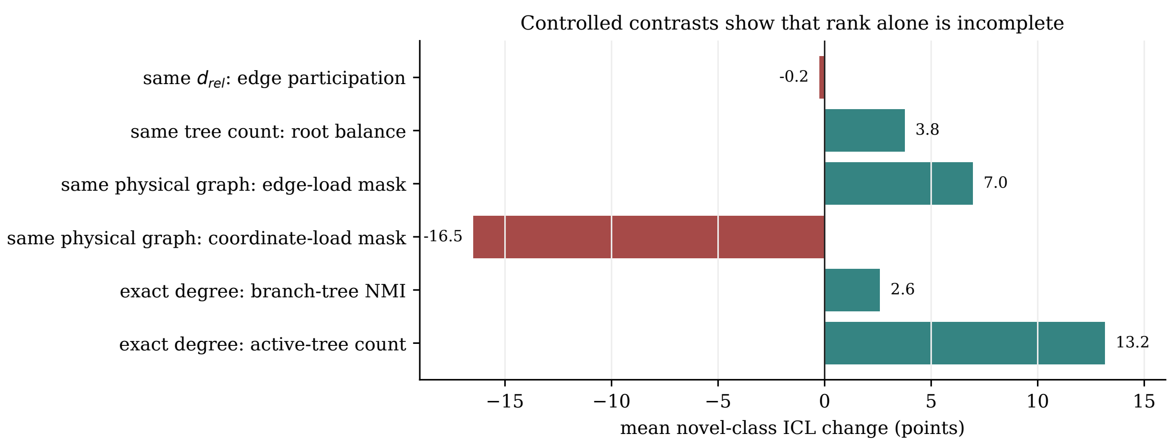

Figure 3: Controlled contrasts. Some coarse structural metrics are not monotone drivers, and fixed-graph input masks can still change ICL. This is why d_{rel} alone is incomplete.

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[7] Post-training mechanism evidence

Pre-training topology asks what the graph could use. Post-training diagnostics ask what the trained model did use. The strongest diagnostic class was projection/root/tree organization around comparison branches. Projection-alignment summaries gave

$$\begin{aligned} R^2_{\mathrm{LOO}} &= 0.740 \quad \text{at run level,} \\ R^2_{\mathrm{LOO}} &= 0.809 \quad \text{at topology-mean level,} \\ R^2_{\mathrm{LOO}} &= 0.599 \quad \text{for topology-best seed.} \end{aligned}$$

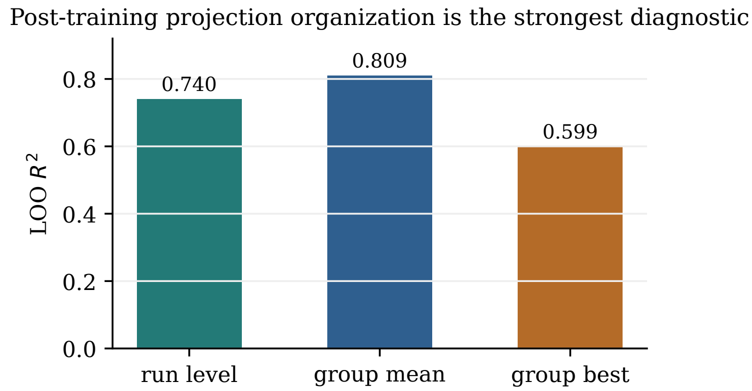

Figure 4: Post-training projection alignment is much more explanatory than coarse pre-training count. This is mechanism evidence, not an independent capacity theorem.

This means high-ICL models are not merely high-count systems. They organize the available tree-sum geometry so that comparison branches produce decodable steady states. The active-tree and margin diagnostics are close to the learned mechanism, so they should be read as mechanistic explanations rather than clean pre-training predictors.

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[8] Causal scrambles

The causal interventions are the strongest evidence that the learned alignment matters. Two statistic-preserving scrambles were central:

branch-alignment scramble	preserves graph, root tree counts, d_{rel}, and input density
projection scramble	also preserves input support and per-edge projection row norms

They break branch/projection/tree organization while preserving many coarse statistics.

In the stratified hard-regime pilot, high-ICL runs dropped by about -96 points under branch-alignment scrambling and about -64 to -68 points under projection scrambling. Low and mid runs also collapsed substantially.

This pushes the evidence beyond correlation. It says the trained CRNs rely on the organized relation between input branches, learned edge projections, and root/tree structure.

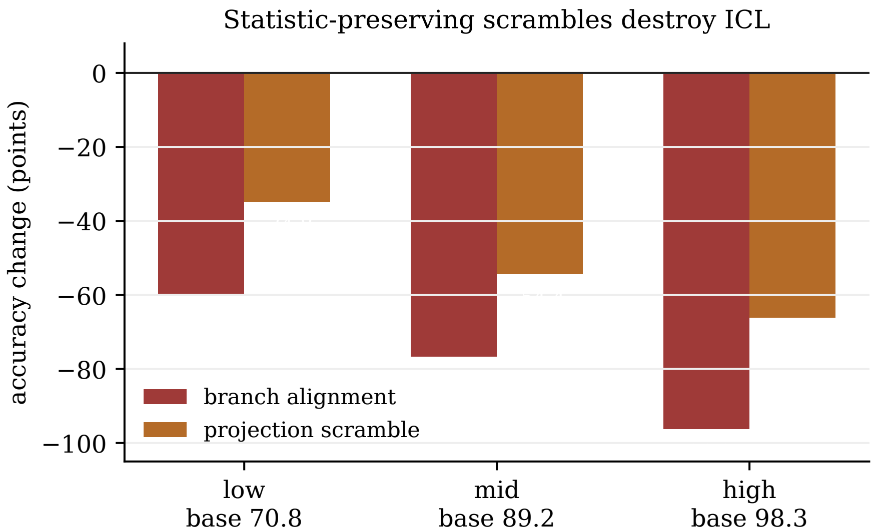

Figure 5: Statistic-preserving scrambles in low, mid, and high baseline buckets. Destroying branch/projection/tree alignment sharply reduces novel-class ICL.

[9] Hard regimes and the normal-fan pilot

Three hard regimes were added: $n4\_m6\_N3\_D2$, $n5\_m8\_N3\_D2$, and $n5\_m12\_N3\_D2$, each with 60 trained runs plus mechanism and causal follow-ups. The results are mixed in the useful sense. Topology signal depends on regime.

In $n5\_m12\_N3\_D2$, masked tree geometry had LOO $R^2 = 0.447$ and positive held-out-family performance. In the sparser $n5\_m8\_N3\_D2$ regime, within-sample bootstrap improvements were more positive than held-out-family prediction. This suggests that topology may have sparse, intermediate, and redundant regimes rather than one monotone scalar rule.

The exact-degree normal-fan pilot fixed

$$N_n = 5, \quad m = 12, \quad N_c = 3, \quad D = 2, \quad \text{in/out degree sequence} = [3, 2, 2, 3, 2], \quad d_{\text{rel}} = 88.$$

It varied tree-polytope/normal-fan diagnostics through directed degree-preserving rewires. Only four topology groups were trained, so this is not yet a statistical result. It is a constructive design showing that topology can vary beyond degree sequence and d_{rel}.

The descriptive $n = 4$ correlations were $r = 0.869$ for active-tree count and $r = 0.691$ for branch-tree NMI with mean ICL. These numbers are promising design feedback, not proof.

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[10] Motifs and matched controls

Essential physical subgraphs retrained better than sparse input masks under the extraction protocol. But matched controls softened the stronger interpretation. Degree-rewired and random strongly connected controls often matched or exceeded the extracted motifs.

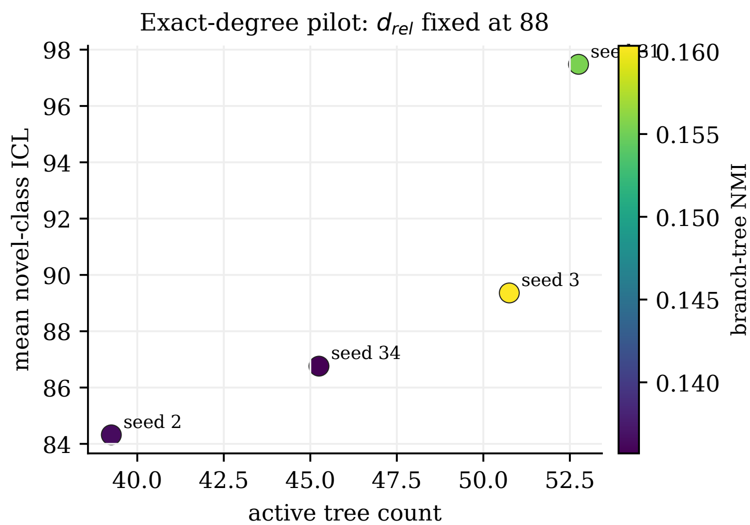

Figure 6: Exact-degree pilot. All points have the same degree sequence and $d_{\mathrm{rel}} = 88$. Active tree count and branch-tree NMI vary across rewires, and mean ICL varies with them in this small $n = 4$ pilot.

The corrected interpretation is:

$\boxed{\text{small physical subgraphs can support ICL, but the extracted motifs are not uniquely optimal.}}$

This is compatible with the tree-sum view: many different graphs can provide usable branch-covering tree geometry.

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[11] The gamma-star attempt

The project implemented γ^*-style capacity probes because d_{rel} is too coarse. This was the right theoretical direction, but the current optimized proxy did not outperform tree geometry. In the hard $n5\_m12\_N3\_D2$ regression, optimized γ capacity had LOO $R^2 = -0.564$, while tree geometry had 0.324 and masked tree geometry had 0.447.

This is an informative failure. The surrogate likely optimizes smooth average behavior more than the true worst-branch margin. ICL is closer to a worst-branch problem:

$$\max_{K,B} \min_{b \in \mathcal{B}} \mathrm{margin}_b(K, B; G, \Omega).$$

A good capacity theory should target low-percentile or worst-branch separation under norm constraints.

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[12] The Markov-ICL extension

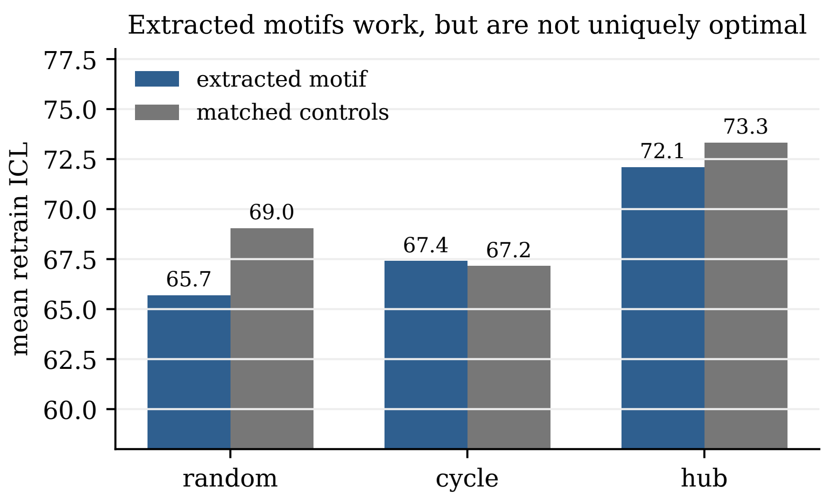

Figure 7: Extracted physical motifs support ICL, but matched controls are competitive. The evidence supports a family of feasible physical substrates, not one unique motif.

The next question was not "run a bigger sweep." It was:

Can the first-order topology story be sharpened using Markov-process expressivity theory?

The requested ingredients were input multiplicity, monotonicity limits, coefficient constraints, non-equilibrium driving, branch sharpness, structural compatibility, and a clean separation between expressivity and trainability.

The exact representation from Section 2 is already the bridge. For first-order CRNs,

$$\tau_r(z) = \sum_{T \in \mathcal{T}_r(G)} \exp(\beta_T + \Theta_T^\top z), \qquad \Theta_T = \sum_{e \in T} K_e.$$

Thus the Markov-expressivity objects must be defined on tree-sum features, not isolated edges.

The input-encoding multiplicity is

$$M_\alpha = \sum_e \Omega_{e\alpha}.$$

For ICL, the important refinements are branch-specific:

$$M_{i,d}, \qquad M_{q,d}, \qquad \sum_e \Omega_{e,i,d}\Omega_{e,q,d}, \qquad |M_{i,d} - M_{q,d}|.$$

Low or imbalanced comparison-coordinate multiplicity is the monotonicity-risk diagnostic:

$$\mathrm{monoRisk}_\alpha = \mathbf{1}[M_\alpha \le 1].$$

After training, an effective participation diagnostic is

$$M_\alpha^{\mathrm{eff}} = \frac{\left(\sum_e |K_{e\alpha}|\right)^2}{\sum_e K_{e\alpha}^2}.$$

Branch sharpness is distinct from branch coverage. For branch direction u_b,

$$R_{r,b} = \max_{T \in \mathcal{T}_r(G)} \Theta_T^\top u_b - \min_{T \in \mathcal{T}_r(G)} \Theta_T^\top u_b.$$

This measures whether a rooted tree polytope can create a sharp margin along a comparison direction. A graph can cover every branch and still have weak lower-tail margins.

Coefficient controllability is the other missing piece. Tree-polynomial coefficients are constrained by overlapping spanning trees; they are not freely adjustable. Therefore rank, even masked rank, can overestimate useful capacity. Effective rank, condition number, tree entropy, normal-fan coverage, extremal-tree accessibility, and branch-specific concentration are all relevant.

The theory note produced in this phase is `ICL/markov_icl_expressivity_theory.md`. Its shortest message is:

> First-order Markov-ICL expressivity is tree-sum expressivity under an input mask. Input multiplicity and branch sharpness are useful structural variables, but not standalone laws.

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[13] Existing-data Markov reanalysis

The reanalysis script `ICL/markov_expressivity_reanalysis.py` consumes committed CSV/JSON artifacts only. It does not launch new broad training. It groups seeds by topology/mask before inference.

Coverage:

dataset	rows	topology/mask groups
fixed m20	240	48
hard $n4\_m6$	60	12
hard $n5\_m8$	60	12
hard $n5\_m12$	60	12
exact-degree pilot	4	4
exact-degree library	32	32

In the fixed m20 data, grouped LOO R^2 for mean novel-class ICL was 0.095 for aggregate multiplicity variables, 0.046 for comparison-multiplicity variables, 0.158 for tree-geometry variables, and 0.145 for the old capacity proxy. In the hard $n5\_m12$ data, tree geometry reached 0.437 while aggregate multiplicity and the old capacity proxy were -0.190.

The input-multiplicity control is especially clarifying. In the fixed-count mask data, all mask families have mean aggregate multiplicity $M_{\text{mean}} = 10$. Yet the coordinate-block family has zero minimum comparison overlap, high coordinate-load Gini, and much lower mean ICL: 63.4 points versus roughly 74–77 points for the other families.

The exact-degree normal-fan pilot was also reinterpreted through this lens. It has only four trained groups, but those groups share the intended fixed-degree and fixed-d_{rel} controls. Descriptive correlations with mean ICL were $r = 0.869$ for normal-fan active tree count, $r = 0.691$ for branch-tree NMI, $r = 0.810$ for log tree count, and $r = 0.759$ for effective rank. These are design signals, not statistical proof.

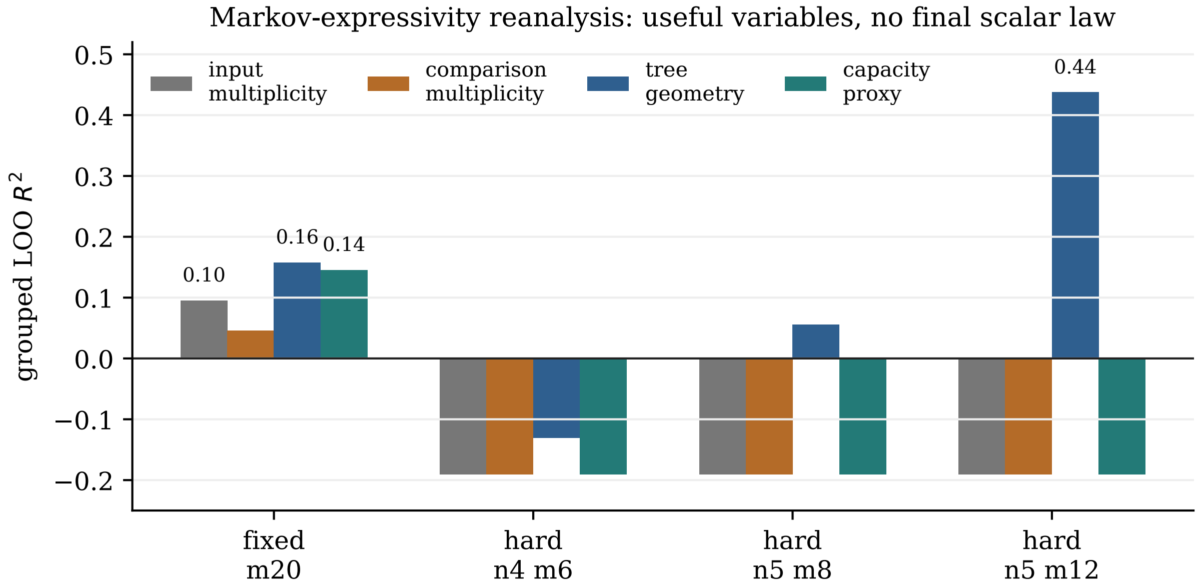

Figure 8: Existing-data Markov reanalysis. The richer Markov variables are useful diagnostics, but the evidence still does not support a universal scalar expressivity law.

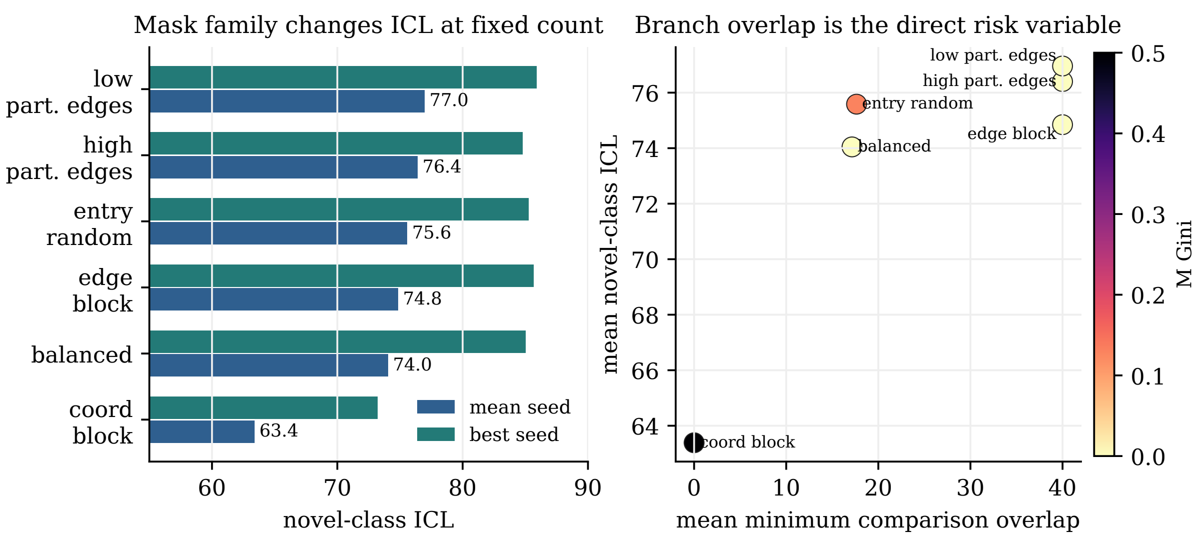

Figure 9: Input multiplicity control from existing data. Average multiplicity alone is not the law; branch comparison overlap and load imbalance are the direct failure risks.

[14] Improved lower-tail capacity

The improved probe is `ICL/branch_margin_capacity_v2.py`. It targets the lower tail directly:

$$\gamma^*_{\mathrm{ICL}}(G, \Omega; \mathcal{C}) = \max_{\theta} \min_{b \in \mathcal{B}} \mathrm{LCVaR}_{\alpha} \left[m_\theta(z) \mid z \in b \right],$$

where

$$m_\theta(z) = q_{y(z)}(z) - \max_{\ell \neq y(z)} q_\ell(z).$$

The implemented variants are exact log-sum-exp tree features, tropical max-over-tree features, and hard-root structural compatibility. The constraints include row-norm bounds for K_e, decoder Frobenius norm bounds, edge-bias bounds, and the input mask Ω.

A small smoke run on a 5-node, 12-edge cycle-chords graph produced finite branch-wise reports for all three variants. The objectives were still negative: -3.001 for exact, -2.800 for tropical, and -2.270 for hard-root. This validates the reporting path and reinforces that average accuracy is not the right target; the worst branch can still fail.

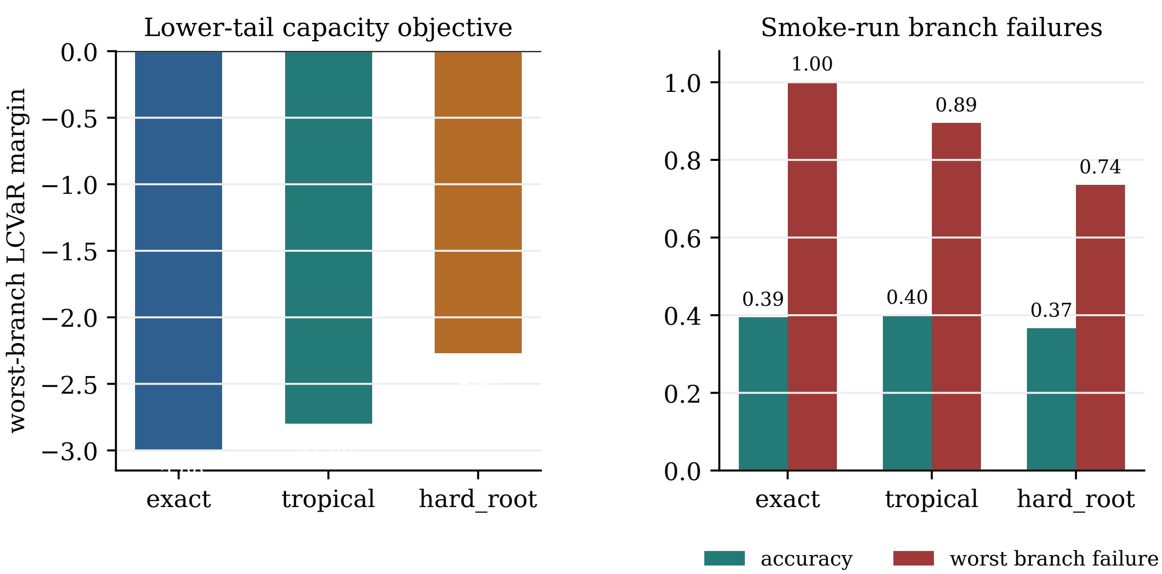

Figure 10: Lower-tail capacity V2 smoke result. The new probe reports the correct object: worst-branch lower-tail margin and branch failures. This smoke run is not an optimized capacity theorem.

[15] Expressivity, trainability, and thermodynamics

The reanalysis keeps three outcomes separate:

$$\text{best seed} \approx \text{expressivity envelope},$$
$$\text{mean seed} \approx \text{trainability/reliability},$$
$$\text{seed variance} \approx \text{optimization instability}.$$

Across the 36 hard-regime groups, masked effective rank correlated 0.895 with best seed and 0.897 with mean seed, but only 0.345 with seed standard deviation. Trained margin and tree entropy were also strongly correlated with best/mean seed, but they are post-training mechanism descriptors rather than pre-training capacity metrics.

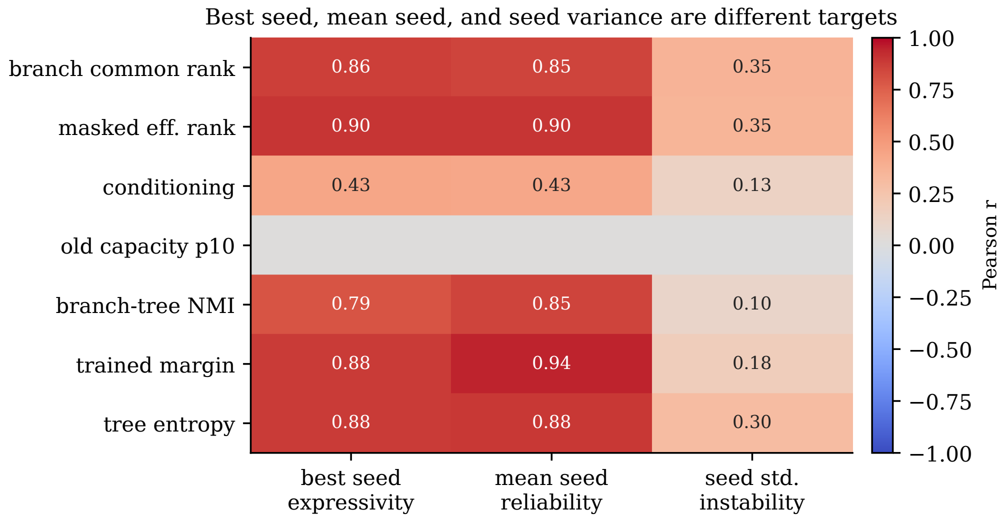

Figure 11: Expressivity and trainability should not be collapsed into one outcome. Best seed, mean seed, and seed standard deviation respond differently to structural and trained-mechanism variables.

The thermodynamic question is deliberately unresolved. Existing arbitrary directed exponential-rate models may be non-equilibrium, but they were not parameterized as reversible-edge thermodynamic Markov processes. Therefore the report `thermodynamic_force_budget_report.md/json` does not claim entropy production or a force-budget law. It only audits eligibility: among 36 local hard topology groups with topology JSON, the mean reversible-edge fraction is 0.512, with range 0 to 1.

A valid thermodynamic experiment needs a reversible support and a parameterization such as

$$W_{ij} = \exp(E_j - B_{ij} + F_{ij}/2 + \text{input drive}), \qquad B_{ij} = B_{ji}, \qquad F_{ij} = -F_{ji}, \qquad |F_{ij}| \le F_{\max}.$$

Only then should we sweep $F_{\max}$ and ask whether ICL branch margins increase with allowed non-equilibrium driving.

[16] What has now been achieved

Before this direction, the paper had:

CRNs can do ICL by subspace projection, and raw learned degrees of freedom are important.

After this direction, the first-order story is sharper:

> raw degrees are useful only insofar as topology and masks turn them into relative tree-sum projections that cover comparison branches.

The concrete accomplishments are:

Layer	What was established
Theory	The exact first-order computational basis is the rooted tree-sum basis $\{\Theta_T\}$, not the isolated edge basis $\{K_e\}$.
Structural evidence	Fixed-count regimes show residual ICL variation explained by tree and masked-tree geometry beyond raw count.
Input topology	Fixed physical graph/count/d_{rel} masks can still differ sharply, so the object is $\gamma^*(G, \Omega)$.
Mechanism	Successful trained models organize projection/root/tree features around comparison branches.
Causality	Statistic-preserving scrambles that break alignment sharply reduce ICL.
Markov expressivity	Input multiplicity, monotonicity risk, branch sharpness, coefficient controllability, and lower-tail capacity were formalized for first-order Markov-ICL.
Trainability	Best seed, mean seed, and seed variance are now treated as different targets rather than one accuracy summary.
Thermodynamics	The current arbitrary directed models were correctly not used for force-budget claims; a reversible-edge $F_{\max}$ sweep is the needed next experiment.
Limits	d_{rel} is not a capacity law; current γ probes are not final; motifs are not unique; nonlinear CRNs are out of scope.

The concrete Markov-extension artifacts added for this phase are:

artifact	purpose
`ICL/markov_icl_expressivity_theory.md`	theory note formalizing tree-sum Markov-ICL expressivity
`ICL/markov_expressivity_reanalysis.py`	existing-data reanalysis and report generator
`ICL/branch_margin_capacity_v2.py`	exact, tropical, and hard-root lower-tail branch-capacity probe
`existing_data_markov_expressivity_reanalysis.md/json`	grouped Markov-expressivity model checks
`input_multiplicity_control_report.md/json`	fixed-count input-mask multiplicity control
`exact_degree_multiplicity_normal_fan_report.md/json`	exact-degree pilot reanalysis
`expressivity_vs_trainability_report.md/json`	best-seed, mean-seed, and seed-variance separation
`thermodynamic_force_budget_report.md/json`	reversible-support audit and valid $F_{\max}$ requirements
`markov_icl_final_synthesis.md`	claim separation: expressivity, trainability, mechanism, thermodynamics
`markov_icl_goal_completion_audit.md`	verification and evidence-boundary audit

The most defensible statement is:

> In the tested first-order fixed-count regimes, topology-associated structural and functional variables explain residual novel-class ICL variation that raw trainable count does not explain. The strongest mechanism evidence is that trained models depend on branch/projection/tree alignment.

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[17] What should come next

The next phase should not be a loose sweep. It should scale the exact-degree/normal-fan design:

$$\text{fix } N_n, m, N_c, D,\ \text{input count, degree sequence, and } d_{\text{rel}},$$
$$\text{vary tree-polytope and normal-fan diagnostics, then train enough groups for grouped inference.}$$

Then compare raw count, d_{rel}, masked tree geometry, normal-fan diagnostics, and an improved worst-branch $\gamma^*(G, \Omega)$ probe.

The theoretical target is:

> derive a norm-constrained tree-polytope branch-margin capacity that predicts whether (G, Ω) can cover ICL branches.

That would turn the current empirical and mechanistic evidence into a true first-order topology theory.